

TICKETS ARE NOT A BUG LIST.

THEY ARE HOW THE FARM RUNS.

Every job at ILCO that needs a human to do something becomes a **ticket**. Daily checks. Cleaning rounds. Equipment calibrations. Mould observations. Driver inspections. Staff requests. RP sign-offs. Owner approvals. SOP executions. SMF reviews. **One inbox. One queue. One source of truth for what needs to happen.**

This wizard walks you through how a ticket is born, lives, and dies — and why making this *the* operational substrate (not a side feature) changes how the entire farm operates.

01 WHY TICKETS-AS-CORE

Three layers, one substrate. Same model for every event.

LAYER 1 · TICKETS

WHAT NEEDS TO HAPPEN

Actionable items with assignee, priority, due time, SLA timer. Includes: SOP execution requests, fault reports, requisitions, RP sign-offs, AR approvals, drift alerts.

LAYER 2 · SPECIALISED DATA

WHAT THE WORK CAPTURED

The rich record of what was actually done — IPM observation values, daily check items, calibration readings, harvest yields, COA results. Each domain has its proper data shape.

LAYER 3 · AUDITLOG

WHAT HAPPENED, IMMUTABLY

Hash-chained, append-only, regulator-grade. Every action by every actor — timestamp, who did it, what they did, hash of the previous entry. Tampering breaks the chain.

WHY THIS DESIGN WINS

Every event in the system — a daily check, a mould sighting, a vaccine due, a calibration overdue, a SAHPRA drift — flows through the same substrate. Staff have one inbox to look at. Auditors have one place to query. Operations leadership has one ticker for facility health. **No more paper SOPs sitting in a binder; no more WhatsApp groups for fault reports; no more "where did we capture that".**

1

TRIGGER

ORIGIN OF THE TICKET

Three things can spawn a ticket: **(a) a person reports it** ("printer broken"), **(b) a scheduled SOP fires** ("daily check Bay 3 · 06:00"), or **(c) the system detects something** ("greenhouse added → SMF C.3.1 stale").

SYSTEM ACTION Ticket row created with priority, category, workflow stage, and either `assignedToId` (a person) or `assignedToRole` (a department).

2

ROUTING

WHO SEES IT

If assigned to a specific person → it appears in their dashboard's **"Mine"** tab. If assigned to a role → it sits in the role's queue and any qualified staff can claim it. RP-bound items land in Berne's queue. Owner-tier items sit in the Owner Approvals banner. Maintenance items go to Calvin's IT queue if it's IT, or to facility maintenance otherwise.

SYSTEM ACTION Routed by role + workflow stage. Notification surfaces appear automatically: ApprovalsWaitingBanner, BottleneckRadar, dashboard widget counts.

3

SLA TIMER KICKS IN

WHEN IT MUST BE ACTED ON

Every ticket has a maximum age before it's "breached". **Critical = 4 hours · High = 24 hours · Medium = 3 days · Low = 1 week**. The ticket card shows a live age stamp; once the SLA is exceeded, it turns red with a 🔥 flame icon.

SYSTEM ACTION No reminders, no nagging emails. The visible flame in the dashboard is the prompt.

4

ACTION — WORK HAPPENS

THE ACTUAL JOB

The assignee opens the ticket, does the work in real life, and captures it in the system. For a daily check: tick the items in the daily-check form. For a mould sighting: photograph + select severity + describe. For a calibration: enter the 3 weight readings. **The ticket form mirrors the paper SOP exactly.**

SYSTEM ACTION A row is written to the relevant specialised table (DailyCheckRecord, IPMObservation, EquipmentCalibration, etc.) — the rich data lives there. The ticket points to it.

5

SIGN-OFF CHAIN (WHERE APPLICABLE)

WHEN REGULATOR-GRADE EVIDENCE IS NEEDED

Some tickets need a sign-off chain before they close — RP_SIGNOFF tickets, COMPLIANCE_APPROVAL, DAR review, AR approval. The chain enforces order: **RP signs first, DAR signs second, AR signs third**. Each signature is timestamped with a real human ID. No skipping, no out-of-order.

SYSTEM ACTION Ticket status moves through the chain. AR approval freezes a snapshot — for SMF sections, this becomes an `SMFSectionVersion` row.

6

RESOLUTION

CLOSING THE LOOP

When the ticket is COMPLETED or CLOSED, the system records `resolvedAt` and `resolvedById` — **the audit trail knows who closed it**. If anyone closes wrongly, a "reopen" path exists with `reopenedById` + `reopenedAt` recorded.

SYSTEM ACTION Closing also clears the ticket from active queues. It still lives in the database — searchable, queryable, never deleted.

7

AUDIT + REFLECTION

FOREVER

Every state change of the ticket is hash-chained into the AuditLog. SAHPRA can later ask "show me every ticket that triggered an SMF change in March" — we run one query and produce the entire trail with full provenance: who, when, what data, what signatures.

SYSTEM ACTION AuditLog entries are append-only and SHA-256 chained. The chain can be verified at any time with `verifyHashChain()` .

03 WHAT THIS REPLACES

Ten things ILCO used to do in 10 different places, now in one.

Paper SOPs in binders	Every SOP execution is a ticket → form mirrors the paper layout → record stored as structured data → audit trail intact
WhatsApp fault reports	Lou photographs mould → opens ticket from his phone → category PEST → routed to Lourens automatically
Email RP sign-offs	Berne sees pending sign-offs in her dashboard queue with priority order; one tap to approve
Spreadsheets for calibration tracking	EquipmentCalibration row spawned by scheduler → due-date enforced → overdue flames red on FM compliance card
Sticky notes for "remind me to..."	Anything is a ticket: requisition, repair, training query, HR request
Manual SAHPRA inspection prep	Auditor pulls AuditLog query → entire chronology with signatures appears
"Did anyone do X?"	Search ticket type + workflow stage → answered in one second
"Who's been overloaded?"	BottleneckRadar surfaces queues ageing past SLA — instantly actionable
Bespoke escalation chains	Workflow stage + priority + role-assignment encode the path; no manual chasing
Drift between SMF + reality	SMF Drift Daemon catches operational changes → raises ticket → Berne acts → re-signs through chain

A sample of the categories live in the system today.

TICKETTYPE	PURPOSE	TYPICAL ASSIGNEE
ISSUE	A fault, observation, or general work item	Person or department
REQUISITION	Supplies, equipment, services request — needs Owner approval	→ Tenant Admin
APPROVAL	Compliance document or operational decision needing sign-off	→ RP / Tenant Admin
RP_SIGNOFF	Batch release, S21 packet, deviation closure — RP-only	→ Berne (RP)
SMF_DRIFT	Operational change invalidated a signed SMF section — auto-raised	→ Berne (RP)
DEEP_CLEAN	Auto-spawned by clearBay — post-harvest deep-clean	→ Housekeeping

ORIGIN'S OPERATING SYSTEM

If a person needs to do something at ILCO, it lives as a ticket. If a regulator needs to review what happened, the audit log replays it. The ticket system is not a bug tracker — it is the operating system of the farm.